

# Micro Integrated Coherent Receiver Application Note

FIM24721, FIM24731, FIM24723, FIM24733 series  
FIM24725, FIM24725 series

---

Fujitsu Optical Components Limited  
February 24<sup>th</sup>, 2021

This document contains Confidential Information which is proprietary to FUJITSU OPTICAL COMPONENTS LIMITED, its subsidiaries and affiliates. No part of its contents may be disclosed or conveyed to, used by or copied for, any third party without the prior written permission of FUJITSU OPTICAL COMPONENTS LIMITED.

## ■ GENERAL SAFETY CONCERNS

**This Receiver module is very sensitive to “Electrical damage”.  
The following procedures are required to work the device safely.**

### ■ Electro-Static Discharge (ESD)

ESD precautions must be taken for AC and DC pins when handling a Integrated receiver. The use of ground straps when using test equipment and test benches that have been properly grounds.

### ■ DC power supply surge protection

Photo-diode and TIA DC pins must be connected the serge suppression filter between all DC power supply.

### ■ Turn off the power while Connecting Operation

DC power must be turned off prior connecting and disconnecting any electrical cables and optical fiber.

### ■ AC power noise protections

All DC power supply grounds and DUT ground must be connected with electrical cables.

Micro Integrated Coherent Receiver (ICR) FIM24721 / FIM24731 / FIM24723 / FIM24733 series are configured with following functions, control pins, monitor pins and power supply pins.

## ■ Functions

- Based on OIF projects
- Eight(8) photo-detectors, comprised of 4 sets of balanced detectors
- Four(4) linear amplifiers with differential outputs
- Two(2) ninety degree hybrid mixers with differential outputs
- A polarization splitter element, separating the input signal into two orthogonal polarizations, with each polarization delivered to a hybrid mixer
- A power splitter, splitting the local oscillator power equally to the two hybrid mixers
- Automatic Gain Control (AGC) and Manual Gain Control (MGC)
- User settable independent output voltage swing adjustment for each of the four outputs
- Peak indicators for each output
- DC blocker for each output
- All above is in one hermetically sealed package
- Variable optical attenuator (VOA)
- Input signal power monitor photodiode (MPD)

## ■ Control pins

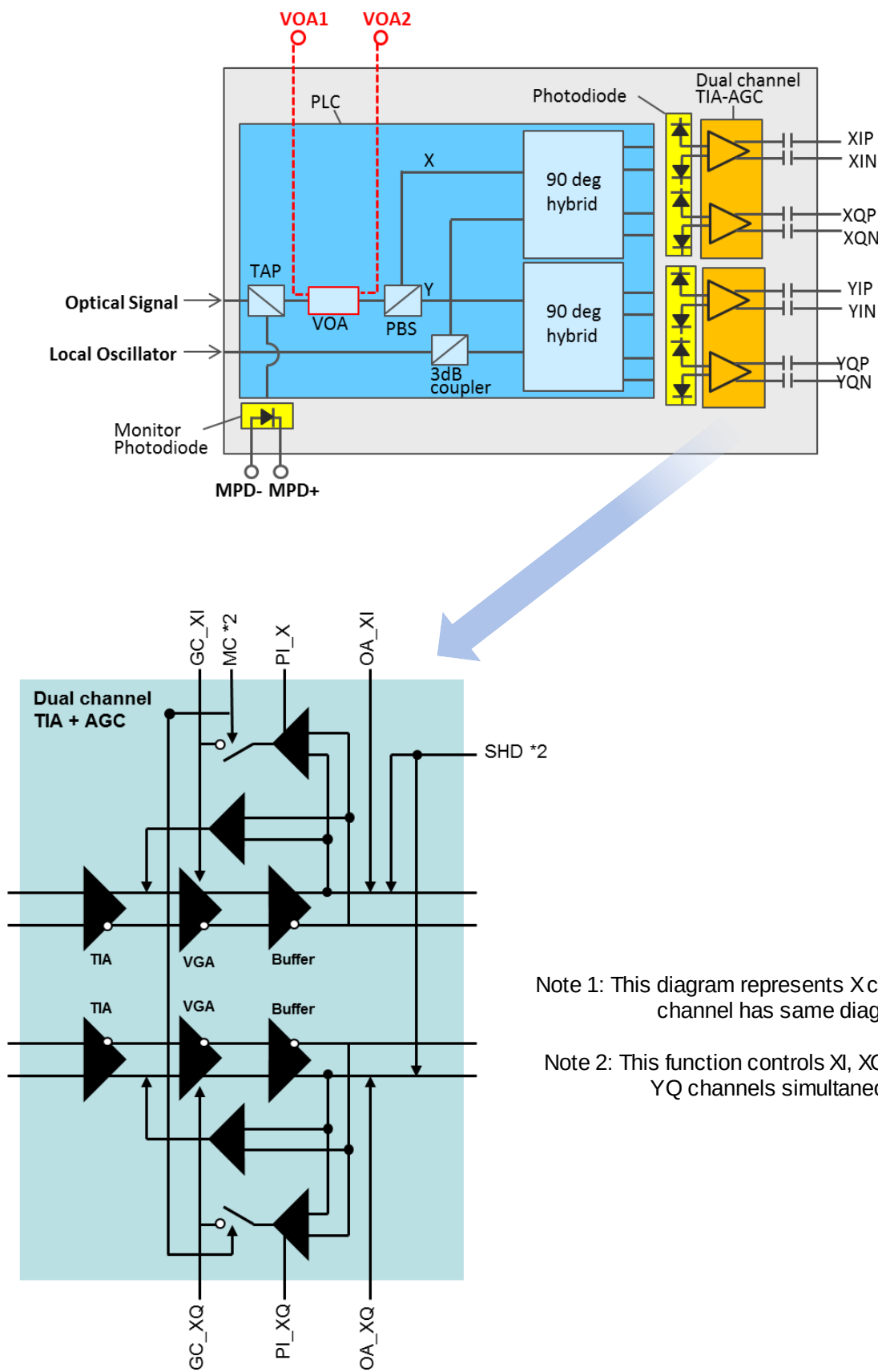
- Gain control mode selection
- Gain adjustment in MGC mode
- Output voltage swing adjustment in AGC mode
- Output shutdown

## ■ Monitor pins

- Peak indicator for output
- Input signal power monitor

## ■ Power supply pins

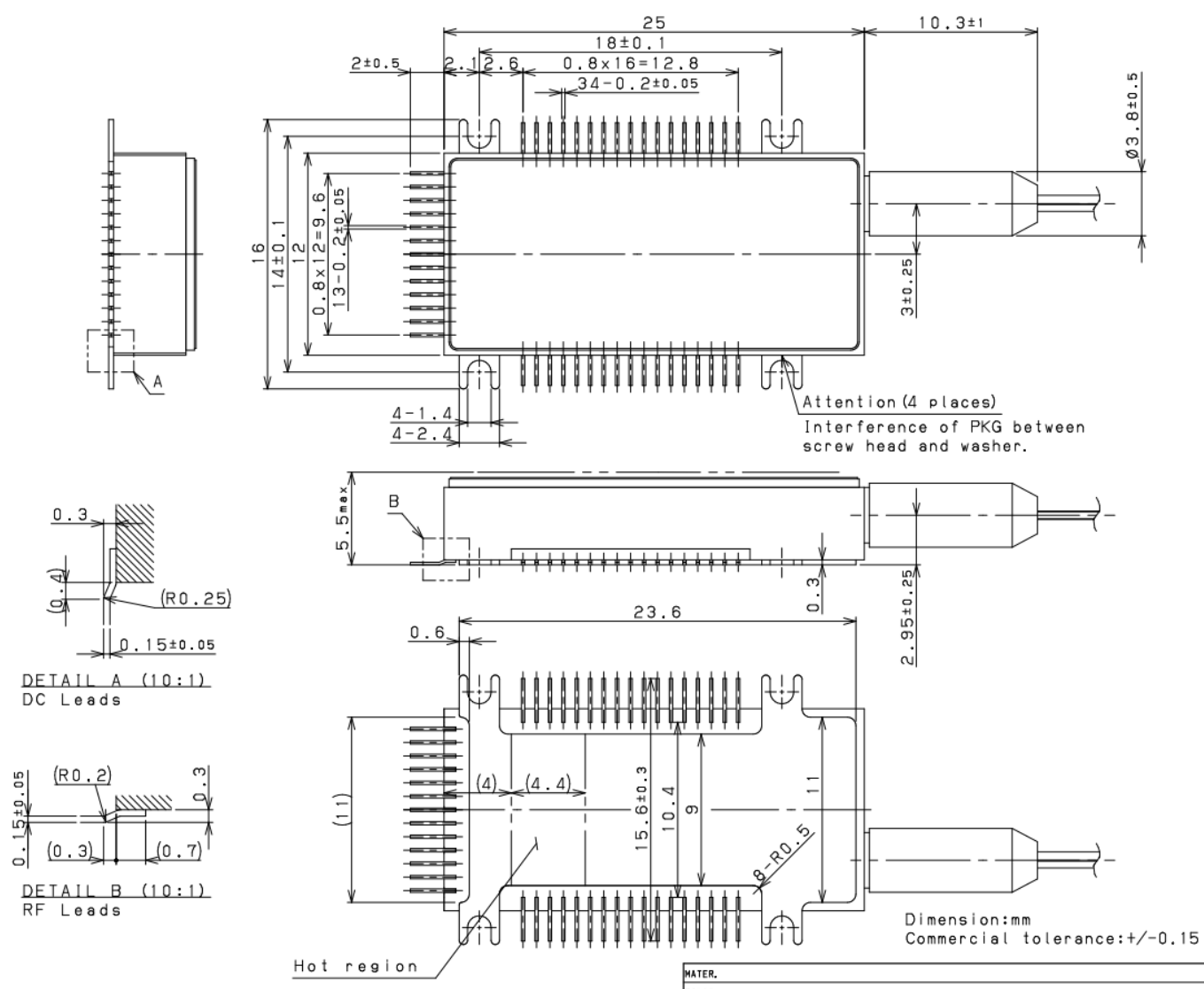
- Photodiode power supply voltages
- Amplifier power supply voltages



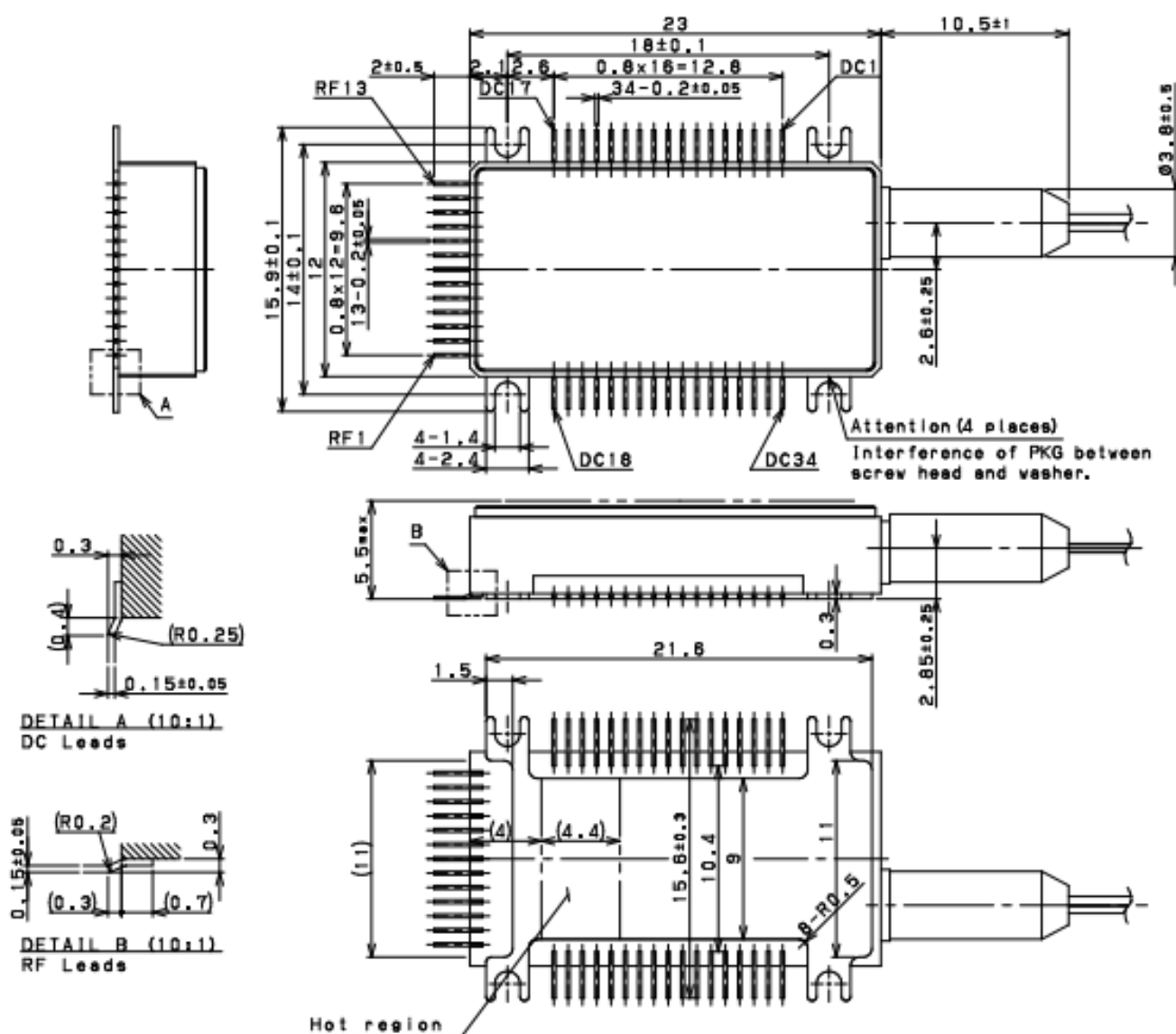
Note 1: This diagram represents X channel. Y channel has same diagram.

Note 2: This function controls XI, XQ, YI and YQ channels simultaneously.

■ Dimensions(FIM24721,FIM24731,FIM24723,FIM24733 series)



### ■ Dimensions (FIM24725,FIM24735 series)



## DC pins

| No | NAME    | FUNCTION                           |
|----|---------|------------------------------------|
| 1  | RFU     | Reserved for Future Use            |
| 2  | RFU     | Reserved for Future Use            |
| 3  | MC/AGC  | MGC/AGC Selection                  |
| 4  | MPD+    | Monitor Photodiode Cathode         |
| 5  | MPD-    | Monitor Photodiode Anode           |
| 6  | VPD_YIn | Photodiode power supply (YIn)      |
| 7  | VPD_YIp | Photodiode power supply (YIp)      |
| 8  | VPD_YQp | Photodiode power supply (YQp)      |
| 9  | VPD_YQn | Photodiode power supply (YQn)      |
| 10 | PI-YI   | Peak Indicator YI                  |
| 11 | GA-YI   | Gain Adjust YI                     |
| 12 | OA-YI   | Output Amplitude Adjust YI         |
| 13 | VCC_Y   | Amplifier power supply (YI and YQ) |
| 14 | GND     | Ground                             |
| 15 | OA-YQ   | Output Amplitude Adjust YQ         |
| 16 | GA-YQ   | Gain Adjust YQ                     |
| 17 | PI-YQ   | Peak Indicator YQ                  |
| 18 | PI-XI   | Peak Indicator XI                  |
| 19 | GA-XI   | Gain Adjust XI                     |
| 20 | OA-XI   | Output Amplitude Adjust XI         |
| 21 | GND     | Ground                             |
| 22 | VCC_X   | Amplifier power supply (XI and XQ) |
| 23 | OA-XQ   | Output Amplitude Adjust XQ         |
| 24 | GA_XQ   | Gain Adjust XQ                     |
| 25 | PI-XQ   | Peak Indicator XQ                  |
| 26 | VPD_XIn | Photodiode power supply (XIn)      |
| 27 | VPD_XIp | Photodiode power supply (XIp)      |
| 28 | VPD_XQp | Photodiode power supply (XQp)      |
| 29 | VPD_XQn | Photodiode power supply (XQn)      |
| 30 | VOA2    | VOA GND                            |
| 31 | VOA1    | VOA control                        |
| 32 | SD      | Shutdown                           |
| 33 | RFU     | Reserved for Future Use            |
| 34 | RFU     | Reserved for Future Use            |

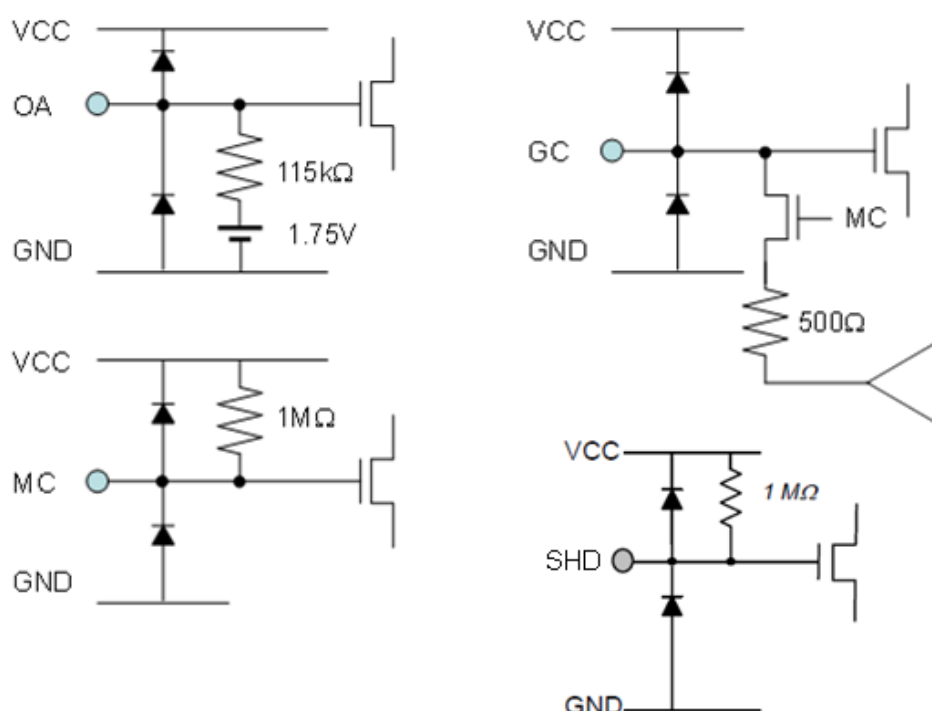
## RF pins

| No | Name |
|----|------|
| 1  | GND  |
| 2  | XIp  |
| 3  | XIn  |
| 4  | GND  |
| 5  | XQp  |
| 6  | XQn  |
| 7  | GND  |
| 8  | YIp  |
| 9  | YIn  |
| 10 | GND  |
| 11 | YQp  |
| 12 | YQn  |
| 13 | GND  |

: Control pins  
 : Monitor pins  
 : Power supply pins

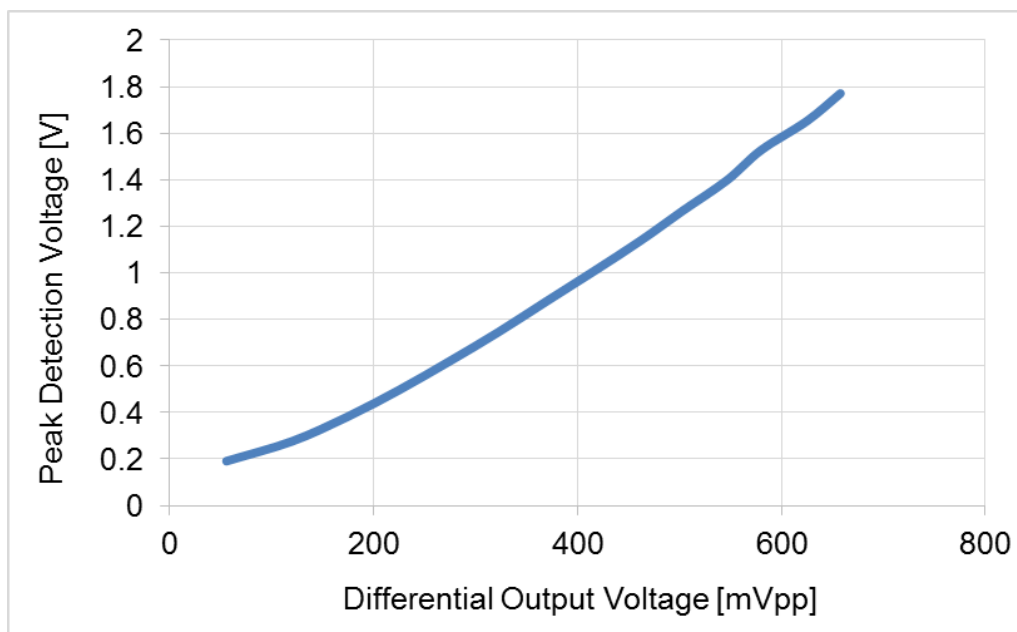
## Control terminals

| Function                                                     |                  | Pin #                | Symbol                           | Min | Typ | Max | Unit |
|--------------------------------------------------------------|------------------|----------------------|----------------------------------|-----|-----|-----|------|
| Gain control mode selection (Digital control)                | MGC mode         | 3                    | MC/AGC                           | 0   | -   | 0.8 | V    |
|                                                              | AGC mode         |                      |                                  | 2   | -   | Vcc |      |
| Gain adjustment in MGC mode (Analog control)                 |                  | 11<br>16<br>19<br>24 | GA_YI<br>GA_YQ<br>GA_XI<br>GA_XQ | 0   | -   | Vcc | V    |
| Output voltage swing adjustment in AGC mode (Analog control) |                  | 12<br>15<br>20<br>23 | OA_YI<br>OA_YQ<br>OA_XI<br>OA_XQ | 0.5 | -   | 2   | V    |
| Output shutdown (Digital control)                            | Shutdown enable  | 32                   | SD                               | 0   | -   | 0.8 | V    |
|                                                              | Shutdown disable |                      |                                  | 2   | -   | Vcc |      |

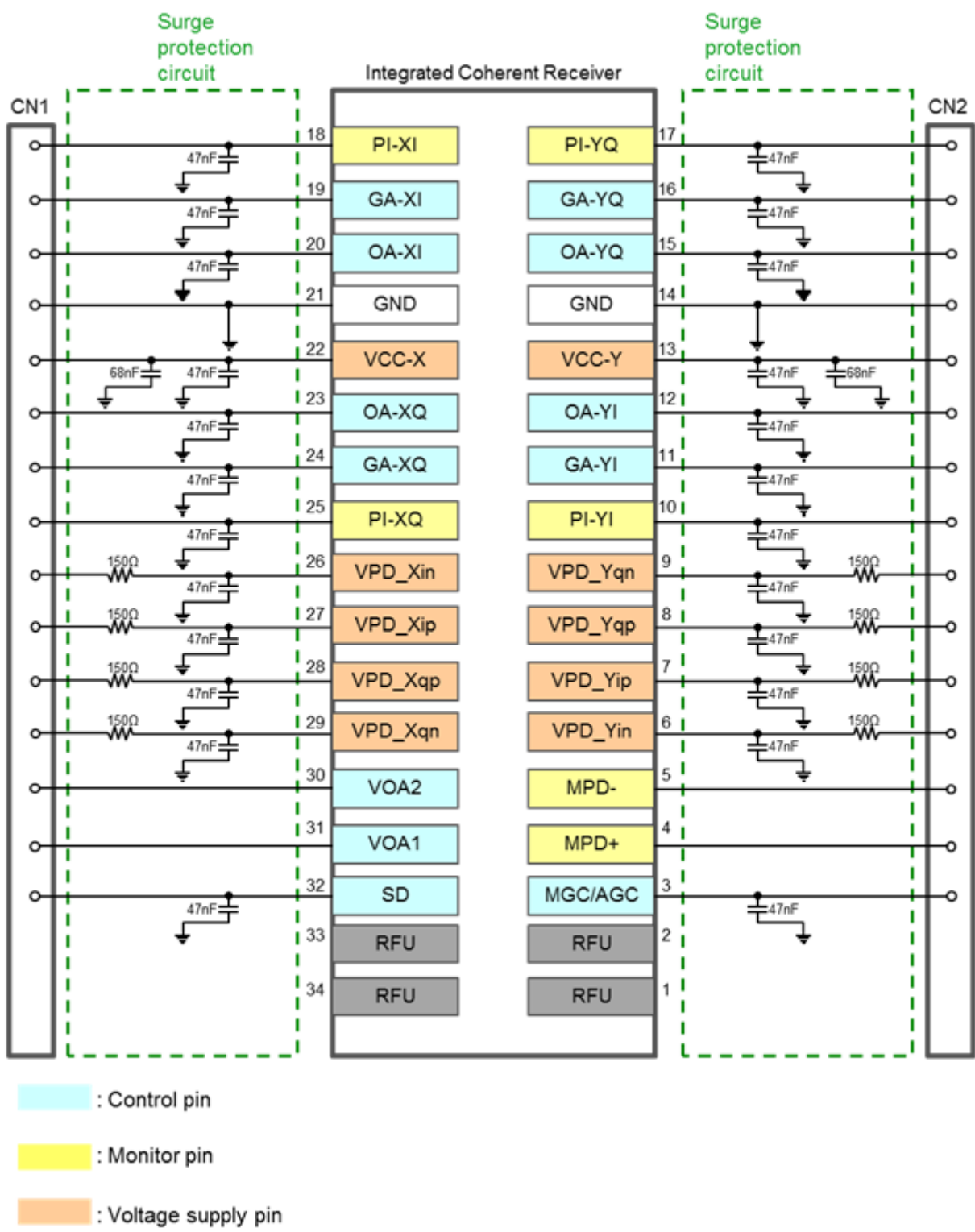


EQUIVALENT CIRCUITS OF CONTROL PINS

## ■ Peak indicator for output



# Recommended electrical circuit of Micro ICR



The Micro ICR should be powered up with the following sequence.

1. Prevent any electrical surge with earth band etc.
2. Confirm the Micro ICR and evaluation board are not connected to any evaluation system. i.e. DC/ AC power supplies/ monitors, optical outputs and other equipment
3. Turn on main switches of DC/ AC power supplies/ monitors.
4. Set DC power supplies to 0 [V]
5. Operate DC power supplies outputs
6. Connect the Micro ICR and evaluation board to evaluation system. i.e. DC/ AC power supply/ monitor and optical input
7. Set photodiode power supplies;  
VPD\_XIp, VPD\_XIn, VPD\_XQp, VPD\_XQn,  
VPD\_YIp, VPD\_YIn, VPD\_YQp, VPD\_YQn,  
from 0 [V] to +5 [V] gradually.
8. Set amplifier power supplies ;  
VCC\_X, VCC\_Y,  
from 0 [V] to +3.3 [V] gradually.
9. Turn on optical input.
10. Adjust control terminal voltages;  
GA\_XI, GA\_XQ, GA\_YI, GA\_YQ,  
OA\_XI, OA\_XQ, OA\_YI, OA\_YQ,  
MC/AGC, SD,  
to get proper performance.

**Note: Step8 can be done simultaneously with step7 though, if amplifier power supplies are up prior to photodiode power supplies, the amplifiers and photodiodes will be seriously damaged.**

The Micro ICR should be powered down with following sequence.

**1. Turn off optical input**

2. Set control terminal voltages to 0[V] gradually.
3. Set amplifier power supplies to 0[V] gradually.
4. Set photodiode power supplies to 0[V] gradually.
5. Disconnect receiver and evaluation board from evaluation system. i.e. DC/AC power supplies/monitors and optical input.
6. Stop to operate DC/AC power supplies/monitors.
7. Turn off main switches of DC/AC power supplies/monitors.

**Note: Step 3 can be done simultaneously with step 2 though, If photodiode power supplies are down prior to amplifier power supplies, the photodiodes and amplifiers will be seriously damaged.**

The Micro ICR includes precision optics. If package of the receiver is deformed by excess load, the Micro ICR may be degraded in its characteristics or damaged.

In order to prevent the degradation or damage, the receiver should be mounted on flat area ( flatness < 15um, surface roughness < 12.5um ) larger than package bottom area including package flanges by M1.2 screws with 0.376kgf\*cm torque.

The Micro ICR is unable to meet aqueous wash requirements.  
FUJITSU OPTICAL COMPONENTS recommends our customers not to wash the Micro ICR in water.  
Not to wash this product in water.  
Not to use this product near a wash bowl, sink or laundry, or in a wet place.



***Note : The ICR in the picture is Example***

## 1. INTRODUCTION

The Evaluation Board is designed to evaluate Micro ICR. For the detail of Micro ICR, please refer each Micro ICR product specification.

## 2. OVERVIEW

The Evaluation Board have the monitor terminals for the analog value from the module and Amplifier power supplies, the jumper switches for terminals control the module behavior. RF outputs with GPPO connectors are 32GBaud electrical signals.

## 3. MONITOR TERMINALS

The terminals for each channel are the following.

| Terminal                           | Silk symbol | Attribution |
|------------------------------------|-------------|-------------|
| Photodiode power supply (**p)      | PD1         | IN          |
| Photodiode power supply (**n)      | PD2         | IN          |
| Peak Indicator                     | PI          | OUT         |
| Amplifier power supply             | VCC_*       | IN          |
| Monitor Photodiode Cathode / Anode | MPD-/+      | IN          |
| VOA1/2 Adjust Voltage              | VOA1/2      | IN          |
| GND                                | GND         | —           |

Note : \* : X or Y

\*\* : XI or XQ or YI or YQ

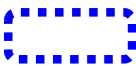
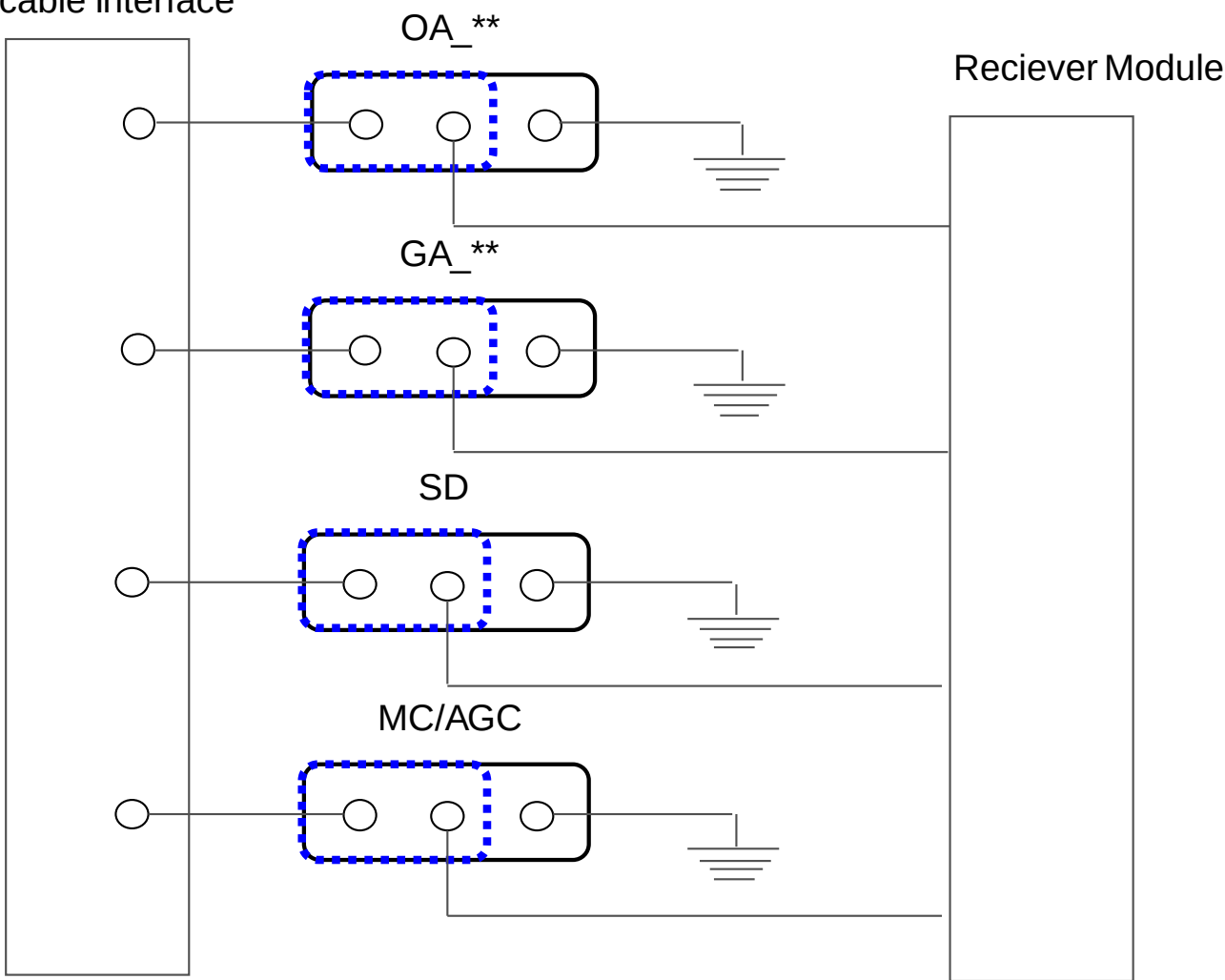
4. CONTROL PINS

The jumper switches for each channel are the following.

| Terminal                   | Silk symbol | Attribution |
|----------------------------|-------------|-------------|
| Output Amplitude Adjust ** | OA_**       | IN          |
| Gain Adjust **             | GA_**       | IN / OUT    |
| Shutdown                   | SD          | IN          |
| MGC/AGC Selection          | MC/AGC      | IN          |

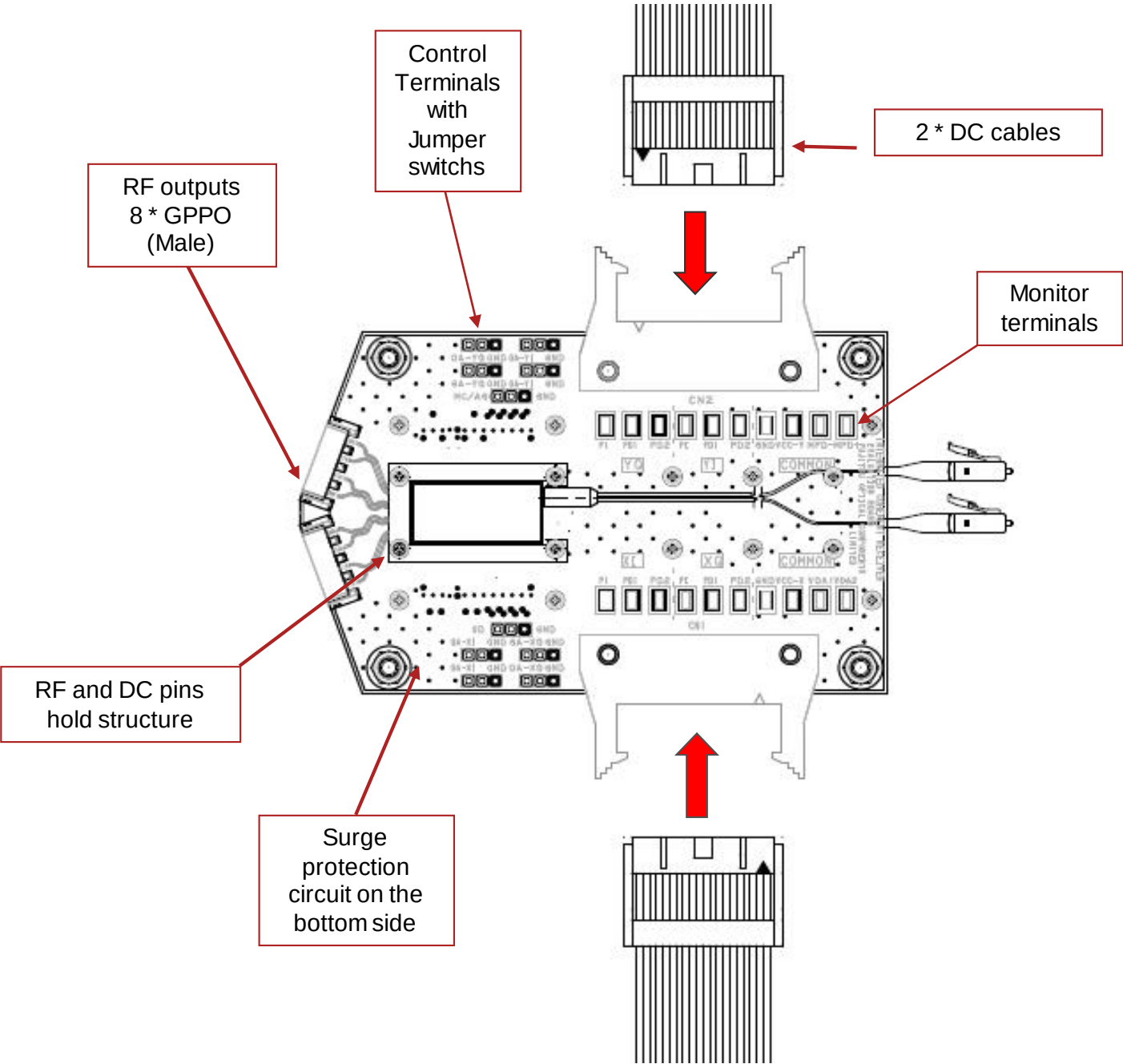
Note : \*\* : XI or XQ or YI or YQ

DC cable Interface



: Default position of the jumper switch

Structure of the evaluation board is shown below.



[VIEW FROM TOP SIDE]

The evaluation board should be set up with following sequence.

1. Prevent any electrical surge with earth band etc.
2. Confirm the Micro ICR is not mounted on the evaluation board.
3. DC cables of the evaluation board to be connected and to be terminated same as Micro ICR pins (refer to "Recommended electrical circuit of Micro ICR"). DC cable numbers correspond to pin numbers of Micro ICR
4. Follow the "Recommended power up sequence of Micro ICR"
5. Follow the "Recommended power down sequence of Micro ICR"

